Kalvakuntla Ritanshu Kumar

9963994865 kalvakuntlaritanshukumar@gmail.com Ritanshu Kumar Ritanshu Kumar Portfolio

Education

Vellore Institute of Technology (VIT) Vellore, India
Integrated M.Tech in Computer Science and Engineering – Data Science ***(CGPA: 8.06)*** Aug 2022 – Sept 2027

Experience

FormWise AI | GitHub | Demo
Independent Developer *2026 – Present*

- Evolved FormWise AI from a Vibe Coding competition prototype into a fully deployed end-to-end AI product, independently rebuilding the Next.js frontend, FastAPI backend, PostgreSQL data layer, and Gemini-powered analysis pipeline.
- Built form creation, publishing, response collection, and analysis workflows, with Gemini generating summaries, sentiment, common themes, key insights, and recommendations from collected responses.
- Deployed the complete application publicly using Vercel, Render, and Supabase, with a documented FastAPI REST API and persistent PostgreSQL storage.

Expense Tracker |, *HTML, CSS, JavaScript, Chart.js* |, GitHub PR
Open Source Contributor

- Implemented a persistent dark-mode feature using CSS custom properties and localStorage, including theme-aware Chart.js labels across the dashboard.
- Tested the feature across the application, resolved a merge conflict, and collaborated with the maintainer through code review; PR was merged.

Projects

MERIT – Market Execution, Risk & Intelligence Testbed |, GitHub |, *Python, Event-Driven Architecture*

- Building an event-driven quantitative trading research platform for studying market-making strategies, execution, inventory management, and risk using realistic market microstructure.
- Designing modular simulation and accounting infrastructure for reproducible experiments, including order-book events, order execution, portfolio state, P&L, and trade-level performance analysis.

RAG Knowledge Assistant |, GitHub |, *Python, ChromaDB, Sentence Transformers, BM25, Cross-Encoder, Groq, Streamlit*

- Built a document-grounded RAG system that recursively ingests PDF, DOCX, and TXT data, using semantic embeddings, BM25, Reciprocal Rank Fusion, and cross-encoder reranking for reliable retrieval.
- Engineered grounded GPT-OSS 20B generation with source citations, refusal handling, conversational follow-ups, streaming responses, and query-level latency instrumentation in a production-style Streamlit interface.

Hybrid CNN-Transformer Model for Pneumonia Detection |, GitHub |, *Python, PyTorch, Flask, Grad-CAM*

- Built HybridXRayNet combining multi-scale CNN blocks, residual attention, Transformer Encoder, adaptive feature fusion, and Grad-CAM for interpretable pneumonia detection.
- Achieved 83.17% accuracy, 79.26% precision, 98.97% recall, 88.03% F1-score, and 96.67% ROC-AUC on a 624-image held-out test set.

CAPTCHA Recognition using CRNN + CTC |, GitHub |, *Python, PyTorch, ResNet18, BiGRU, CTC*

- Developed an end-to-end CAPTCHA recognition system combining ResNet18 visual feature extraction, bidirectional GRU sequence modeling, and CTC decoding without explicit character segmentation.
- Trained and evaluated on 9,955 CAPTCHA images, achieving 99.30% exact-string accuracy and 99.86% character-level accuracy on a 1,991-image held-out test set.

Technical Skills

Languages: Python, SQL, C++, Java, JavaScript, TypeScript
AI/ML: PyTorch, TensorFlow/Keras, Scikit-learn, CNNs, Transformers, NLP, Computer Vision, LLMs, RAG, Embeddings
Backend & Data: FastAPI, Flask, Streamlit, React.js, Next.js, PostgreSQL, SQLite, ChromaDB, SQLAlchemy, REST APIs
Quantitative & Research: Pandas, NumPy, Statistics, Market Microstructure, Event-Driven Simulation, Quantitative Research
Tools & Infrastructure: Git, GitHub, Docker, pytest, Ruff, mypy, AWS, Vercel, Render, Supabase

Certifications

100x Dev: Live Full Stack Open Source Cohort-1
Forage: Tata GenAI Powered Data Analytics Job Simulation
Coursera: Supervised Machine Learning
Embrizon: Machine Learning with Python